Relationship Between Additional Employment Stability Fee and Foreign Workers in Taiwan

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1. Introduction

〈1〉Research Background

In 1989, Taiwan established a system for importing foreign laborers, marking the first introduction of foreign workers to engage in low-skilled jobs. With Taiwan's population aging, the demand for caregiving services has steadily increased. Starting in 1992, Taiwan began allowing foreign caregivers and domestic helpers to work in the country. By the year 2022, statistics from the Directorate-General of Budget, Accounting and Statistics of the Executive Yuan showed that the majority of Taiwan's foreign workers hail from Southeast Asian countries such as the Philippines, Indonesia, and Vietnam. They are primarily employed in manufacturing and caregiving industries.

The introduction of foreign labor has undoubtedly had a significant impact on Taiwan's labor market, providing substantial labor supply in a relatively short period. Wage regulations and various policies have influenced the labor market from multiple perspectives, indirectly altering the structure of Taiwan's labor market.

Distinct cultural and language differences have given rise to complex issues, including integration with employers, assessment of foreign workers' rights, and the applicability of Taiwan's Labor Standards Act. These issues have spurred ongoing discussions within Taiwan, reflecting the high degree of attention and concern surrounding the topic domestically.

〈2〉Research Objectives

While numerous studies have traditionally emphasized the impact of introducing foreign labor on the domestic workforce, this study aims to shift attention towards the effects of Taiwan's government policies on foreign laborers themselves.

This research article specifically investigates the implications of the Employment Stability Fee and Additional Employment Stability Fee that employers are required to pay for foreign workers. The study seeks to understand how these policies influence the wages of foreign laborers and the mechanisms through which such impacts manifest in the labor market.

Key objectives include:

1. **Impact on Foreign Laborer Wages:** Analyzing how the implementation of ESF and AESF policies affects the wage levels of foreign workers in Taiwan.
2. **Labor Market Responses:** Exploring the reactions and adjustments within the labor market in response to these policy measures, including potential shifts in employment patterns, labor demand, and employer behaviors towards foreign workers.

By focusing on these aspects, the study aims to provide insights into the direct effects of Taiwan's regulatory framework on foreign laborers, contributing to a nuanced understanding of policy impacts within the broader context of labor market dynamics.

2.Policy and Economic Theory

〈1〉Policies Regarding Foreign Migrant Workers

Before delving into the topic of Employment Stability Fee, it's important to briefly introduce the "3K Five-Level System" in the manufacturing industry, as it bears significant relevance to policies concerning foreign migrant workers. Implemented by the government in 2010, this system applies to 105 specified industries characterized by abnormal temperatures, dust, toxic gasses, organic solvents, chemical treatments, non-automated operations, and other designated processes.

These industries are categorized into five levels (A+, A, B, C, D) based on their specific production characteristics. One of the major distinctions among these levels is the allocation ratio for foreign migrant workers, which varies as follows: A+ (35%), A (25%), B (20%), C (15%), and D (10%).

	A+	A	B	C	D
highest rate	35%	25%	20%	15%	10%

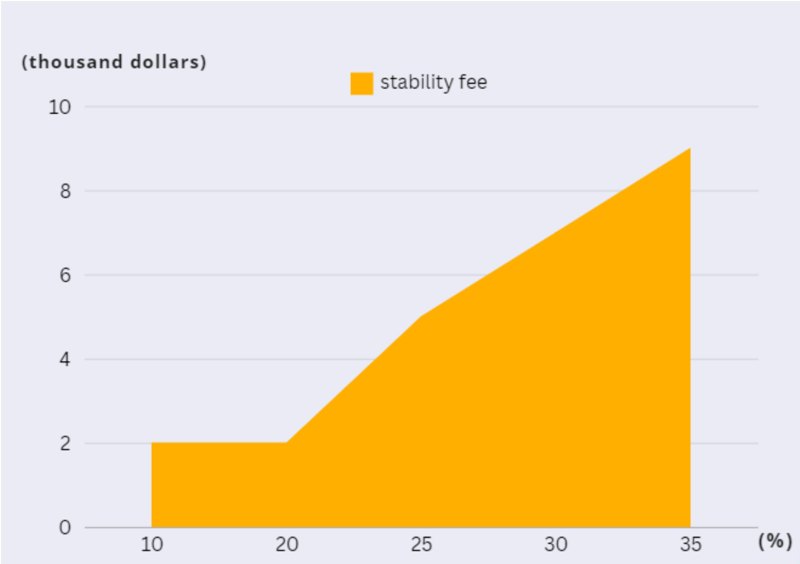
This categorization directly influences the hiring and deployment of foreign workers within these industries, reflecting the government's strategy to manage and regulate labor utilization based on industry-specific risks and requirements. Understanding this system is crucial for contextualizing how policies like the Employment Stability Fee interact with regulations governing foreign labor in Taiwan's manufacturing sector.

Starting from 2013, Taiwan implemented an additional mechanism known as the Employment Stability Fee, which depends on the "3K Five-Level System" allocation rates. Employers are required to pay an extra fee if they exceed the maximum allowable allocation rate for their industry grade when hiring foreign workers. This policy directly influences decisions regarding foreign labor employment and contributes significantly to labor cost management within these sectors.

extra rate	*original fee*	*extra fee*	*total*
5%	$2000	$3000	$5000
5%-10%	$2000	$5000	$7000
10%-15%	$2000	$7000	$9000

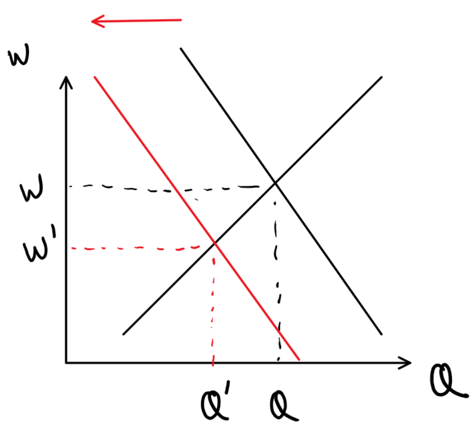
The additional employment allocation rate can go up to an extra 20%, with the exception of the trading industry, which can go up to 40%. For example, if it belongs to the B-grade industry and wishes to employ 30% foreign workers, exceeding the maximum allocation rate by 10%, then for each additional foreign worker hired, a monthly additional Employment Stability Fee of 5,000 yuan is required (this 5,000 yuan includes the original stability fund of 2,000 yuan).

Below is a chart illustrating the total Employment Stability Fee, including the original 2,000 yuan and additional ESF, for industries with a maximum allocation rate of 20% when employing different percentages of foreign workers (unit: thousand yuan).

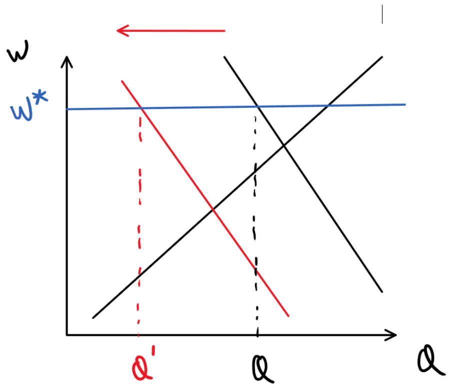


〈2〉Labor Market Supply and Demand

For employers, the Employment Stability Fee represents an additional cost they must pay to the government when hiring foreign workers, thereby increasing the overall cost of employing them. Consequently, it is anticipated that employers may reduce their hiring of foreign migrant workers. This would shift the demand curve for foreign labor to the left, resulting in a leftward movement of the equilibrium point of supply and demand. Both the wage and the number of foreign workers employed would decrease as illustrated below:



However, considering the impact of the minimum wage, the foreign labor market is not at the equilibrium point of perfect competition. If the employment stabilization fee causes a leftward shift of the demand curve, it may not necessarily lead to a decrease in wages, but rather a decrease in the number of employees, as shown in the diagram below:



3.Research Methods and Steps

〈1〉Source of Data

The empirical data for this study is derived from the Ministry of Labor's "Survey on the Management and Utilization of Foreign Workers." The survey targets employers approved by the Ministry of Labor's Workforce Development Agency for employing migrant workers in business and household sectors. The scope of the survey covers manufacturing, construction, healthcare and social work services, and other service industries in Taiwan, including Kinmen and Matsu counties.

Two types of survey forms are used: the "Business Sector Migrant Worker Survey Form" and the "Foreign Household Care Worker Survey Form." The sampling method employed is stratified proportional sampling. For the business sector, variables such as industry and company size are used for stratification, while for the household sector, variables such as county/city and nationality are utilized.

The survey is conducted via mail with telephone follow-ups. The study utilizes data from the business sector spanning from the year 2011 to 2019, excluding the 2014 "Survey on Foreign Workers' Work and Living Conditions" due to the absence of relevant variables needed for the study. The total sample size ranges from 4,451 to 5,839 annually, amounting to a total of 40,737 samples over eight years.

〈2〉Research Subjects

Due to time and resource constraints, this study focuses specifically on certain industries within the business sector data. The selection method involves first calculating the proportion of factories applying for the employment stabilization fee across different industries. Industries with the highest and lowest application rates are selected, followed by two additional industries with a higher volume of data available.

(Note: The calculation is based on data from the years 2015 to 2019, as the variable regarding application for the employment stabilization fee was not included in the data from 2011 to 2014.)

Industry Codes	Sample Sizes (Years 2015-2019)	Number of Applications	Percentage of Applications
4	**314**	**80**	**0.255**
9	35	9	0.257
11	344	107	0.311
5	191	60	0.314

12	100	32	0.320
13	473	153	0.323
21	**4010**	**1358**	**0.339**
20	534	181	0.339
19	203	73	0.360
8	480	176	0.367
2	100	38	0.380
16	828	318	0.384
23	552	214	0.388
17	**7751**	**3013**	**0.389**
25	562	224	0.399
15	839	348	0.415
22	730	303	0.415
18	698	293	0.420
14	2361	1006	0.426
10	251	107	0.426
3	1275	547	0.429
1	1605	699	0.436
24	415	181	0.436
7	**519**	**252**	**0.486**
26	99	*NA*	

Final Selection Results:

1. Textiles and Garments Manufacturing Industry (Industry Code 4), Sample Size: 499.
2. Machinery and Equipment Manufacturing Industry (Industry Code 21), Sample Size: 6,417.
3. Metal Products Manufacturing Industry (Industry Code 17), Sample Size: 11,584.
4. Pulp, Paper, and Paper Products Manufacturing Industry (Industry Code 7), Sample Size: 833.

〈3〉Definitions and Measurement of Variables

(1) Dependent Variable: Wage

Based on the survey data, the dependent variable "Wage" is defined as the average total monthly earnings per foreign worker, which includes regular wages and overtime pay, surveyed each June for each respective year.

(2) Independent Variables

Employment Stabilization Fee: Since the data does not directly provide the additional employment stabilization fee paid by firms, this study calculates the average stabilization fee per foreign worker for each firm. This calculation is based on the proportion of foreign workers employed by each firm, government regulations regarding the additional stabilization fee, and the legally mandated issuance rate for each industry.

$$Average\ Additional\ Employment\ Stabilization\ Fee\ per\ Foreign\ Worker = \frac{\text{廠商為所有外籍移工支付之外加安定費總額}}{\text{外籍移工雇傭量}}$$

$$= \frac{\text{超過法定核發率5\%內之移工數*3000+超過法定核發率5−10\%之移工數*5000+...}}{\text{外籍移工雇傭量}}$$

To simplify data processing, an approximate calculation method is used as follows:
(Pre-calculate perc = Percentage of foreign workers in each factory)

$$Average\ Additional\ Employment\ Stabilization\ Fee\ per\ Employee =$$

$$\begin{cases} 0 & \text{if } perc \leq 0.2 \\ 3000 \times (perc - 0.2) & \text{if } 0.2 < perc \leq 0.25 \\ 150 + 2000 \times (perc - 0.25) & \text{if } 0.25 < perc \leq 0.3 \\ 250 + 2000 \times (perc - 0.3) & \text{if } 0.3 < perc \leq 0.35 \\ 350 + 2000 \times (perc - 0.35) & \text{if } 0.35 < perc \leq 0.4 \end{cases}$$

$$Average\ Additional\ Employment\ Stabilization\ Fee\ per\ Foreign\ Worker = \frac{\text{平均每名員工外加安定費}}{perc}$$

(3) Control Variables

- Factory Size: Since the data does not provide specific categories for factory size that might fully capture its impact on the dependent variable, this study will use total number of employees as a proxy for factory size.
- Industry Type: Industry type will be used as a control variable in different regression models by selecting specific industries rather than adding dummy variables for each industry.
- Year: Year will be included as a control variable by adding dummy variables for each year in the regression models.

〈3〉Analysis Method

ChatGBuilding Four Multiple Regression Models for Different Industries to Examine the Impact of Additional Employment Stabilization Fee on Foreign Workers' Wages, with Descriptive Statistical Analysis

4.Analysis of Research Results

Here are the results of the regression analysis:

	Textiles and Garments Manufacturing Industry			Pulp, Paper, and Paper Products Manufacturing Industry		
Additional Employment Stabilization Fee	**0.3101**	**(0.438)**	***	**0.20**	**(0.29)**	
Factory Size	2.69	(0.7593)		11.98	(1.18)	***
year_101	1604	(537.5)	**	234.97	(450.10)	
year_102	545.4	(594.5)		-408.03	(491.02)	
year_104	1594	(533.1)	**	99.38	(432.99)	
year_105	1573	(524.5)	***	651.69	(443.12)	
year_106	2043	(571.4)	***	1191.85	(452.96)	**
year_107	2865	(571.7)	***	1720.81	(450.26)	***
year_108	3685	(560)	***	2963.20	(456.00)	***
constant	21530	(398.9)		22903.71	(341.64)	***

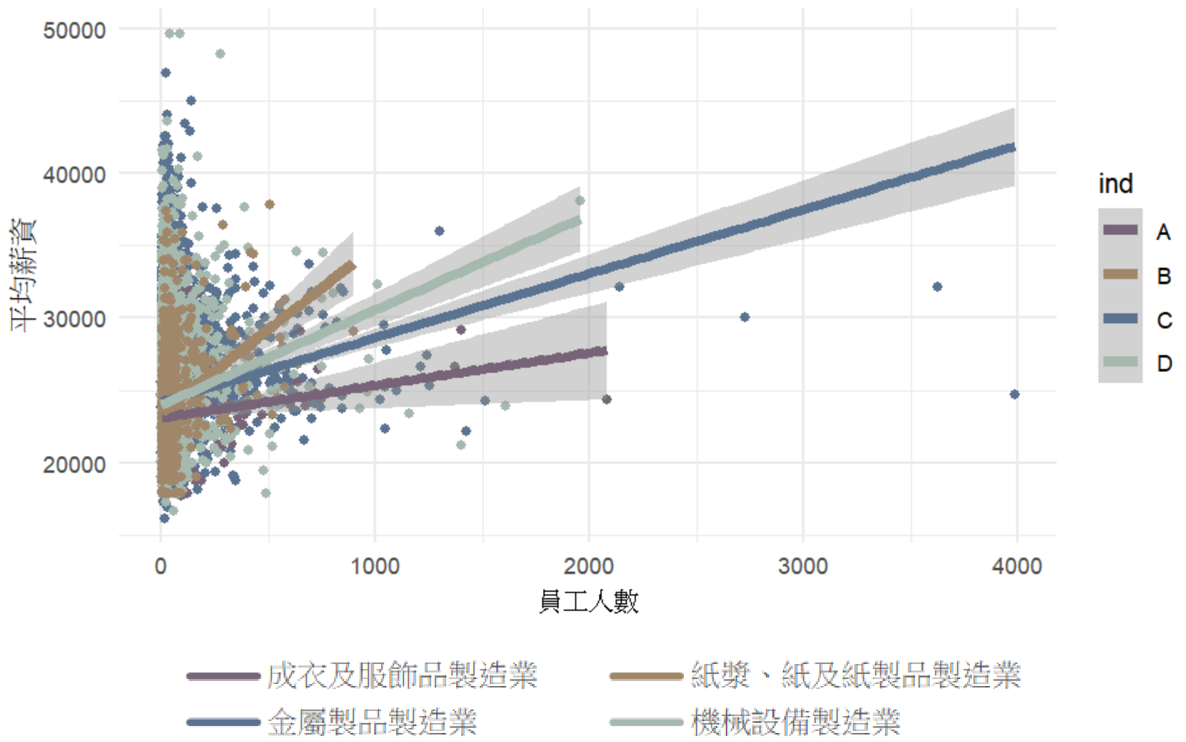
Adjusted R-squared	0.1277	0.1728
p-value	4.499E-12	< 2.2e-16
N	499	833

	Metal Products Manufacturing Industry			Machinery and Equipment Manufacturing Industry		
Additional Employment Stabilization Fee	**0.4961**	**0.09034**	***	**-0.28**	**(0.14)**	*
FactorySize	4.223	(0.3055)	***	6.14	(0.54)	***
year_101	1065	(136.6)	***	936.83	(174.68)	***
year_102	298.1	(146.3)	*	172.72	(193.37)	
year_104	1559	(129.6)	***	1425.74	(178.72)	***
year_105	1367	(128.7)	***	1461.14	(174.09)	***
year_106	2038	(133.1)	***	2129.70	(179.24)	***
year_107	2439	(135.7)	***	2949.75	(185.52)	***
year_108	3388	(134.4)	***	3551.54	(184.31)	***
constant	22790	(101.9)	***	22728.68	(135.83)	***
Adjusted R-squared	0.09683			0.1127		
p-value	< 2.2e-16			< 2.2e-16		
N	11584			6417		

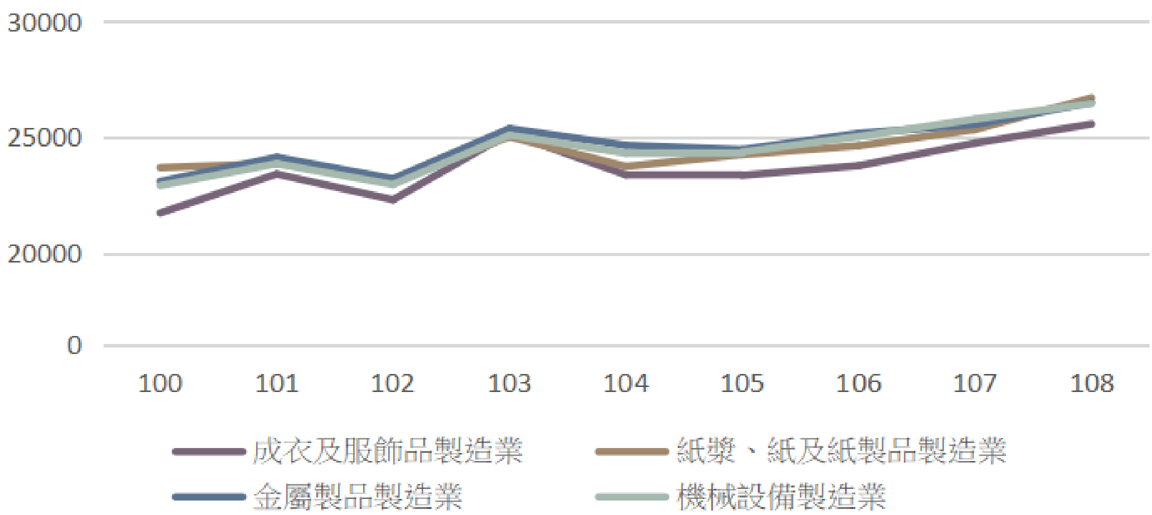
〈3〉The impact of fundamental variables on wages

In this study, four industries were selected, and four regression models were constructed. The impact of variables on wages varies across these models, and I will explain them sequentially.

Regarding the variable of factory size, observed in the Textiles and Garments Manufacturing Industry (referred to as the garment industry here), this variable did not show a significant impact on wages. However, it was highly significant in the other three industries. This suggests that in these three industries, larger factories tend to pay higher wages to foreign workers.



Next, regarding the time variable, the study found that wages generally increased each year except for the year 2013, which saw a decline across all industries due to adjustments in the additional stabilization fee. The Pulp, Paper, and Paper Products Manufacturing Industry showed non-significant growth from 2012 to 2016, possibly influenced by external factors or due to a smaller sample size.



〈2〉The impact of the stabilization fee on wages

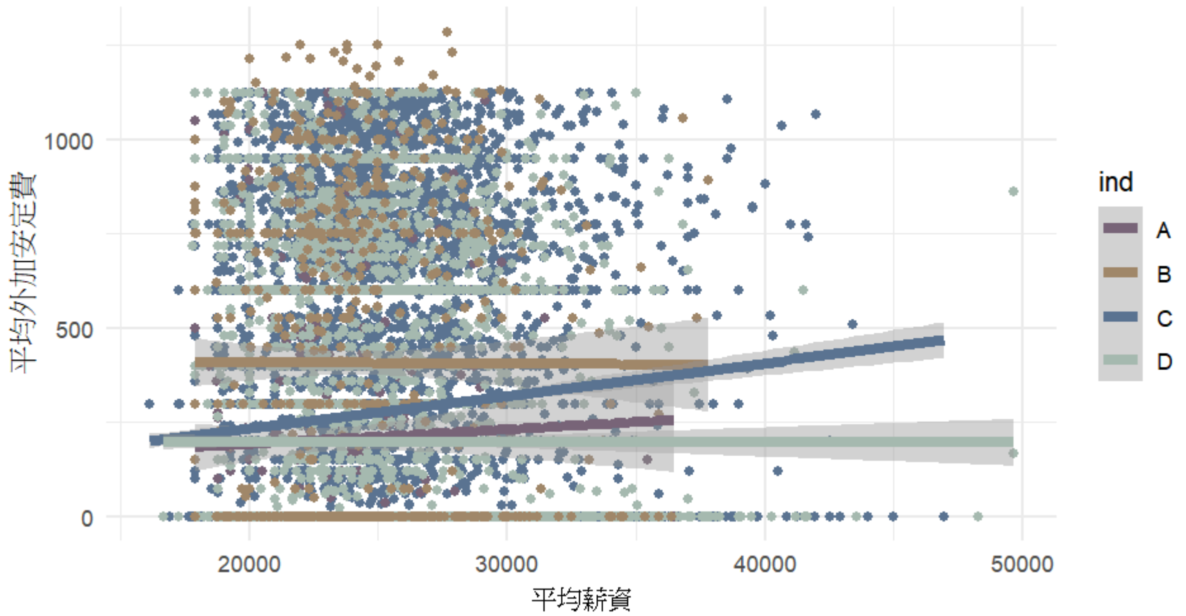
Following the observation of the impact of the stabilization fee on wages, it was found that in the Textiles and Garments Manufacturing Industry and the Metal Products Manufacturing Industry, the stabilization fee had a highly significant positive correlation with wages. In contrast, the Paper Products Manufacturing Industry showed no significant correlation, while the Machinery Manufacturing Industry exhibited a significant negative correlation. This contrasts with our initial predictions, prompting further investigation into potential reasons.

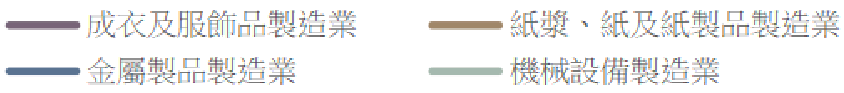
According to Chen Kunming's study titled "Reanalysis of the Impact of Introducing Foreign Labor on the Taiwanese Economy," Taiwan's real wages exhibit a significant degree of downward rigidity. The influence of foreign workers on domestic wages is limited. Similarly, when stabilization fees are higher, firms employing foreign workers may not lower their wages accordingly. This could partially explain why a negative correlation was not observed, though it is not the primary reason.

Firms that pay higher stabilization fees likely have a higher proportion of foreign workers, indicating a stronger demand for foreign labor. Consequently, these firms are more inclined to hire foreign workers at higher wages. This endogeneity factor may be the primary cause of these results. Moreover, it can be inferred that this endogenous effect is stronger in the Textiles and Garments Manufacturing Industry and the Metal Products Manufacturing Industry, where firms experiencing greater labor shortages are more likely to offer higher wages. In contrast, this endogenous relationship appears weaker in the Machinery Manufacturing Industry.

〈3〉Impact Across Different Industries

In different industries, the impact of employment stabilization fees on wages varies, but the exact reasons remain unclear based on current research findings. From the graph, it can be observed that the only industry with a negative slope is Machinery Manufacturing (D), where the overall average of additional stabilization fees is relatively low, likely due to lower issuance rates in this sector. In the Metal Products Manufacturing industry (C), there is a noticeable trend where firms facing labor shortages tend to pay higher stabilization fees. This could be attributed to significant differences in value-added among firms in this industry, allowing more profitable firms to hire more foreign workers.





5.Conclusion and Recommendations

〈1〉Firms that pay higher employment stabilization fees are generally willing to offer higher wages to attract labor.

Firms that need to pay higher employment stabilization fees often employ a higher proportion of foreign workers, indicating their preference to replace local workers with foreign ones. This could stem from high local wages or inadequate supply of local labor. From a tax perspective, these firms, with lower demand elasticity, can generate more tax revenue, aligning with tax efficiency. However, efforts to reduce foreign labor hiring to create job opportunities for locals may be less effective. To effectively reduce foreign labor numbers, a uniform tax rate based on employment ratios could incentivize firms with higher demand elasticity to hire local workers instead.

〈2〉The negative impact of stabilization fees on wages is typically smaller than the endogenous effect caused by labor shortages within firms.

From the perspective of firms, the decision to hire labor depends on marginal costs and marginal output. While the marginal cost of hiring foreign workers increases due to higher stabilization fees, firms with higher marginal output will continue to hire workers at higher wages. In contrast, firms with lower marginal output will cease hiring foreign workers, thereby avoiding the need to pay stabilization fees.

〈3〉The study does not provide insights into the overall market impact of changes in stabilization fee rates.

Because this study focuses on the relationship between stabilization fees and wages paid by different firms during the same period, and the firms' responses have already concluded, our analysis based on current conditions cannot eliminate the characteristics of different firms, nor can it exclude the most significant endogenous factors. Such analysis can only differentiate between different firms or industries and cannot observe the overall market response when fee rates change.

〈4〉Unable to determine the impact of stabilization fees on the hiring levels of foreign workers.

The data source for this study lacks annual continuity in samples, thus preventing an understanding of how firms adjust the hiring levels of foreign workers when stabilization fees change.

6.References

Ministry of Labor, "Survey on the Management and Utilization of Foreign Workers"

Kun-Ming Chen, "Reanalysis of the Economic Impact of Introducing Foreign Workers to Taiwan"

Directorate-General of Budget, Accounting and Statistics, Executive Yuan (Taiwan), data sources